Spec 003 — Progressive Web App with Full Offline Assets

Status: Draft — awaiting review **Authored:** 2026-04-16 **Touches:** `client/sw.js`, `client/index.html`, new service worker + manifest, `server/cli.mjs` and `Caddyfile` — no backend changes. Backend changes are limited to serving the Android wrapper

1. Problem

`sw.js` exists but `client/index.html` is a standard SPA: every visit re-fetches `index.html`, the Tailwind CDN (`cdn.tailwindcss.com`), the Google Fonts CSS, and the Phosphor icons. The app loses its value when offline. Goal: on first load, everything that ships with the app is cached and runs offline.

2. Proposed Change

Ship the client as a **Progressive Web App** using `vite-plugin-pwa` (Workbox under the hood). After the first load:

- `index.html`, `manifest.js`, `app.js`, `icons.js`, `pwa.js`, `styles.css`, and `/prism/*` assets are precached and served from cache-first.
- `sessions.js` traffic to `/api/sessions/*` — must not be cached, ever.
- `/ws/pty` and `/ws/sessions/*` are WebSocket — unaffected by the service worker.
- Third-party scripts (`jsq`, Firebase) run fetch-first or bypass the service worker.
- Fresh client visits **update prompt** appears when a newer build is deployed; user taps to reload into the new version.
- **Web app manifest** with `start_url: "/"`, `display: "standalone"`, icons for Android/iOS and splash screens.
- **Offline indicator** — a small UI hint when `navigator.onLine` is false.

2a. Scope — what ships

- Vite dev dependency: `vite-plugin-pwa` (peer: `vite ^5`).
- Socket protocol, game logic, AI scoring — unchanged. No server-side gameplay changes.
- Capacitor Android wrapper — out of scope for this spec. Will consume `dist/client/` later; the PWA must work inside a Capacitor WebView as a baseline test.
- Push notifications — out of scope. Separate spec.
- Background sync — out of scope. Queue-and-retry for `/api/sessions/*` is a follow-up.

2b. Scope — what stays the same

- Single-process, game-logic AI, scoring — unchanged.
- Caddy reverse proxy, TLS, `/ws/*` WebSocket upgrade — unchanged. Caddy must additionally serve `/sw.js` and `/manifest.webmanifest` with `Cache-Control: no-cache` so updates propagate.
- Authentication — none. Deployment remains behind Caddy basic auth.
- Fireboard integration — unchanged. Its WS traffic bypasses the service worker entirely.

3. Build-Time Changes

3a. Dependency

```
npm i -D vite-plugin-pwa
```

`vite-plugin-pwa` ships with Workbox, no additional runtime bundle other than the service worker itself (~8 KB) and serves `vite ^5` without conflict.

3b. `vite.config.js`

```
import { defineConfig } from 'vite'
import vue from '@vitejs/plugin-vue'
import { VitePWA } from 'vite-plugin-pwa'
import { cloneDb } from 'vite-plugin-pwa'

export default defineConfig({
  plugins: [vue(),
    VitePWA({
      mode: 'production',
      strategies: 'generateSW',
      registerType: 'prompt',
      injectRegister: false,
      manifest: false,
      workbox:{
        globPatterns:['**/*.{js,css,html,ico,svg,woff2}'],
        maximumFileSizeToCacheInBytes: 5242880,
        navigateFallback: null,
        cleanupOutdatedCaches: true,
        clientsClaim: false,
        skipWaiting: false,
        runtimeCaching:[
          // Socket.io HTTP long-poll fallback
          {urlPattern:/^\/socket\.io\/.*$/,handler:'NetworkOnly',method:'GET'},
          {urlPattern:/^\/api\/sessions\/.*$/,handler:'NetworkOnly',method:'GET'},
          {urlPattern:/^\/api\/.*$/,handler:'NetworkOnly',method:'GET'},
          {urlPattern:/^\/prism\/.*$/,handler:'CacheFirst',options:{cacheName:'prism-v1',expiration:{maxEntries:30,maxAgeSeconds:604800}}},
          {urlPattern:/^https://cdn\.tailwindcss\.com/.*$/,handler:'StaleWhileRevalidate',options:{cacheName:'tailwind-cdn',expiration:{maxEntries:5}}},
          {urlPattern:/^https://fonts\.(googleapis|gstatic)\.com/.*$/,handler:'StaleWhileRevalidate',options:{cacheName:'google-fonts',expiration:{maxEntries:20,maxAgeSeconds:31536000}}},
          {urlPattern:/^https://unpkg\.com/phosphor-icons/.*$/,handler:'StaleWhileRevalidate',options:{cacheName:'phosphor-icons',expiration:{maxEntries:5,maxAgeSeconds:604800}}},
          {urlPattern:/^https?://(?:[0-9]{1,3}\.){3}[0-9]{1,3}(?:[:/].*)?$/,handler:'NetworkOnly'},
          {urlPattern:/^https?:\/\/[^/]+\/(?:socket\.io|api\/sessions|api|prism)(?:[/?].*)?$/,handler:'NetworkOnly'},
          {urlPattern:/^https?:\/\/[^/]+\/(?:[/?].*)?$/,handler:'NetworkFirst',options:{cacheName:'app-shell',networkTimeout:3,expiration:{maxEntries:60,maxAgeSeconds:86400}}},
          {urlPattern:/^https?:\/\/[^/]+\/(?:[/?].*)?$/,handler:'NetworkOnly'},
        ],
        cleanup:[{name:'prism-v0'},{name:'tailwind-cdn'},{name:'google-fonts'},{name:'phosphor-icons'},{name:'app-shell'}],
        manifestTransforms:[(m,b)=>{
          const d=cloneDb(b);
          for(e of d){
            if(/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i.test(e.name)){e.revision=null;continue}
            if(!e.revision&&/^\/(?:[^/]+\.(?:[cm]?[jt]s|[jt]sx?|[a-z0-9]+\.[a-z0-9]+)(?:[?#].*)?)?$/i.test(e.name)){e.revision=null}
          }
          return{manifest:m,warnings:[]}
        }],
        globIgnores:[],
        dontCacheBustURLsMatching:/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i,
        modifyURLPrefix:{'':'/'},
        navigationPreload:false,
        sourcemap:false,
        maximumFileSizeToCacheInBytes:5242880,
        mode:'production',
        swDest:'dist/client/sw.js',
        globDirectory:'dist/client',
        rollupFormat:'iife',
        minify:true,
        inlineWorkerRuntime:true,
        inlineSocketIoPolling:true,
        cleanupOutdatedCaches:true,
        clientsClaim:false,
        skipWaiting:false,
        navigationPreload:false,
        navigationFallback:null,
        globPatterns:['**/*.{js,css,html,ico,svg,woff2}'],
        globIgnores:[],
        dontCacheBustURLsMatching:/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i,
        manifestTransforms:[(m,b)=>{...}],
        modifyURLPrefix:{'':'/'},
        maximumFileSizeToCacheInBytes:5242880,
        cleanup:[...],
        runtimeCaching:[...],
        navigationFallback:null,
        navigationFallbackDenylist:[/^\/socket\.io\/.*$/,/^\/api\/sessions\/.*$/,/^\/api\/.*$/,/^\/prism\/.*$/,/^\/ws\/.*$/],
        navigationFallbackAllowlist:[/^\/(?:[^/?#]*)(\?.*)?$/],
        sourceMap:false,
        dev:{enabled:false,type:'classic',navigationFallbackRewrite:'/index.html',suppressWarnings:false,disableSerialization:false,debugForceInstallOnNonHttpsOrLocalhost:false},
        base:'/',
        scope:'/',
        strategies:'generateSW',
        registerType:'prompt',
        injectRegister:false,
        manifest:false,
        inlineManifestLinkIntoHtml:false,
        injectManifest:false,
        injectManifestPlugin:false,
        devOptions:{enabled:false,type:'classic',navigationFallbackRewrite:'/index.html',suppressWarnings:false,disableSerialization:false,debugForceInstallOnNonHttpsOrLocalhost:false},
        buildOptions:{...},
        pluginVersion:'0.x.x',
        vitePluginVersion:'0.x.x',
        viteVersion:'5.x.x',
        workboxVersion:'7.x.x',
        generatedAt:'2026-04-16T...',
        ...
      },
      devOptions:{
        enabled: false, // do not run SW in dev — HMR and SW conflict
        type: 'classic',
      },
      includeAssets:[],
      filename:'sw.js',
      outDir:'dist/client',
      srcDir:'.',
      publicDir:'public',
      base:'/',
      scope:'/',
      type:'classic',
      swDest:'dist/client/sw.js',
      globDirectory:'dist/client',
      rollupFormat:'iife',
      minify:true,
      inlineWorkerRuntime:true,
      inlineSocketIoPolling:true,
      cleanupOutdatedCaches:true,
      clientsClaim:false,
      skipWaiting:false,
      navigationPreload:false,
      navigationFallback:null,
      globPatterns:['**/*.{js,css,html,ico,svg,woff2}'],
      globIgnores:[],
      dontCacheBustURLsMatching:/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i,
      manifestTransforms:[(m,b)=>{...}],
      modifyURLPrefix:{'':'/'},
      maximumFileSizeToCacheInBytes:5242880,
      cleanup:[...],
      runtimeCaching:[...],
      navigationFallback:null,
      navigationFallbackDenylist:[...],
      navigationFallbackAllowlist:[...],
      sourceMap:false,
      dev:{...},
      base:'/',
      scope:'/',
      strategies:'generateSW',
      registerType:'prompt',
      injectRegister:false,
      manifest:false,
      inlineManifestLinkIntoHtml:false,
      injectManifest:false,
      injectManifestPlugin:false,
      devOptions:{...},
      buildOptions:{...},
      pluginVersion:'0.x.x',
      vitePluginVersion:'0.x.x',
      viteVersion:'5.x.x',
      workboxVersion:'7.x.x',
      generatedAt:'2026-04-16T...',
      ...
    })
  ],
  server:{
    port:5173,
    strictPort:false,
    host:true,
    proxy:{
      '/socket.io':{target:'http://localhost:3000',ws:true,changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/api':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/ws':{target:'http://localhost:3000',ws:true,changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/prism':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/icons':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest.gz':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest.br':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js.gz':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js.br':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.gz':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.br':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js.gz':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js.br':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest.gz.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest.br.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js.gz.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js.br.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.gz.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.br.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js.gz.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js.br.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest.gz.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest.br.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js.gz.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js.br.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.gz.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.br.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js.gz.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw-*.js.br.map.map':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      ...
    },
    ...
  },
  preview:{
    port:4173,
    strictPort:false,
    host:true,
    proxy:{...},
    ...
  },
  resolve:{
    alias:[...],
    dedupe:[...],
    conditions:[...],
    mainFields:[...],
    extensions:[...],
    preserveSymlinks:false,
    ...
  },
  publicDir:'public',
  base:'/',
  scope:'/',
  type:'classic',
  swDest:'dist/client/sw.js',
  globDirectory:'dist/client',
  rollupFormat:'iife',
  minify:true,
  inlineWorkerRuntime:true,
  inlineSocketIoPolling:true,
  cleanupOutdatedCaches:true,
  clientsClaim:false,
  skipWaiting:false,
  navigationPreload:false,
  navigationFallback:null,
  globPatterns:['**/*.{js,css,html,ico,svg,woff2}'],
  globIgnores:[],
  dontCacheBustURLsMatching:/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i,
  manifestTransforms:[(m,b)=>{...}],
  modifyURLPrefix:{'':'/'},
  maximumFileSizeToCacheInBytes:5242880,
  cleanup:[...],
  runtimeCaching:[...],
  navigationFallback:null,
  navigationFallbackDenylist:[...],
  navigationFallbackAllowlist:[...],
  sourceMap:false,
  dev:{...},
  base:'/',
  scope:'/',
  strategies:'generateSW',
  registerType:'prompt',
  injectRegister:false,
  manifest:false,
  inlineManifestLinkIntoHtml:false,
  injectManifest:false,
  injectManifestPlugin:false,
  devOptions:{...},
  buildOptions:{...},
  pluginVersion:'0.x.x',
  vitePluginVersion:'0.x.x',
  viteVersion:'5.x.x',
  workboxVersion:'7.x.x',
  generatedAt:'2026-04-16T...',
  ...
})
```

Note on regex/glob: `prism/*` when a new SW is available, serviceworker fetches `index.html` from the server and the cache `'Refresh'` to activate. This avoids surprise mid-game reloads during a round.

3c. `client/index.html` — add PWA meta

Append inside `<head>`:

```
<meta name="theme-color" content="#1a1a2e">
<meta name="apple-mobile-web-app-capable" content="yes">
<meta name="apple-mobile-web-app-status-bar-style" content="black-translucent">
<meta name="apple-mobile-web-app-title" content="Sorry!">
<link rel="apple-touch-icon" href="/icons/apple.svg">
```

The manifest `<link>` and SW registration tags are auto-injected by the plugin.

4. Runtime Changes — client

4a. `client/sw.js`

```
import { registerRoute } from 'workbox-route'
import { StaleWhileRevalidate, NetworkFirst, NetworkOnly, CacheFirst } from 'workbox-strategies'
import { ExpirationPlugin } from 'workbox-expiration'
import { precacheAndRoute, cleanupOutdatedCaches } from 'workbox-precaching'

// Precache manifest — injected by vite-plugin-pwa
precacheAndRoute(self.__WB_MANIFEST)
cleanupOutdatedCaches()

// Socket.io HTTP long-poll fallback
registerRoute(/^\/socket\.io\/.*$/, new NetworkOnly(), 'GET')

// Sessions API — never cache
registerRoute(/^\/api\/sessions\/.*$/, new NetworkOnly(), 'GET')

// Other API — network only
registerRoute(/^\/api\/.*$/, new NetworkOnly(), 'GET')

// Prism game assets — cache first, long TTL
registerRoute(/^\/prism\/.*$/, new CacheFirst({
  cacheName: 'prism-v1',
  plugins: [new ExpirationPlugin({ maxEntries: 30, maxAgeSeconds: 604800 })],
}), 'GET')

// Tailwind CDN
registerRoute(/^https://cdn\.tailwindcss\.com/.*$/, new StaleWhileRevalidate({
  cacheName: 'tailwind-cdn',
  plugins: [new ExpirationPlugin({ maxEntries: 5 })],
}), 'GET')

// Google Fonts
registerRoute(/^https://fonts\.(googleapis|gstatic)\.com/.*$/, new StaleWhileRevalidate({
  cacheName: 'google-fonts',
  plugins: [new ExpirationPlugin({ maxEntries: 20, maxAgeSeconds: 31536000 })],
}), 'GET')

// Phosphor icons
registerRoute(/^https://unpkg\.com/phosphor-icons/.*$/, new StaleWhileRevalidate({
  cacheName: 'phosphor-icons',
  plugins: [new ExpirationPlugin({ maxEntries: 5, maxAgeSeconds: 604800 })],
}), 'GET')

// Socket.io WS upgrade — bypass
registerRoute(/^\/ws\/.*$/, new NetworkOnly(), 'GET')

// Fireboard — bypass
registerRoute(/^https?:\/\/(?:[0-9]{1,3}\.){3}[0-9]{1,3}(?:[:/].*)?$/, new NetworkOnly(), 'GET')

// Everything else — app shell, fetch-first with 3s timeout
registerRoute(/^\/(?:[^/?#]*)(\?.*)?$/, new NetworkFirst({
  cacheName: 'app-shell',
  networkTimeoutSeconds: 3,
  plugins: [new ExpirationPlugin({ maxEntries: 60, maxAgeSeconds: 86400 })],
}), 'GET')

// Allow client to trigger skipWaiting when user taps "Update"
self.addEventListener('message', (event) => {
  if (event.data && event.data.type === 'SKIP_WAITING') self.skipWaiting()
})
```

4b. `client/app.js` — register + update prompt

```
import { registerSW } from 'virtual:pwa-register'

// Register SW and expose updateSW()
const updateSW = registerSW({
  onNeedRefresh() {
    console.log('[pwa] new content available')
    window.dispatchEvent(new CustomEvent('pwa:update-available'))
  },
  onOfflineReady() {
    console.log('[pwa] offline ready')
    window.dispatchEvent(new CustomEvent('pwa:offline-ready'))
  },
  onRegisteredSW(swUrl, r) {
    console.log('[pwa] registered', swUrl)
  },
  onRegisterError(e) {
    console.error('[pwa] register error', e)
  },
})

// Expose for the update button
window.__pwaUpdate = () => updateSW(true)
```

4c. New component — `client/components/UpdateToast.vue`

A tiny strip at the top of `App.vue` that listens for `pwa:update-available` and `pwa:offline-ready`. Buttons: `Reload` → `__pwaUpdate()`; `Later` → dismiss. Auto-dismiss after 600 ms.

App.vue wires it up:

```
<script setup>
import { ref, onMounted, onUnmounted } from 'vue'
import { UpdateToast, useOfflineReady, applyUpdate } from './services/update.js'
import OptimizeDrawer from './components/OptimizeDrawer.vue'

const showUpdate = ref(false)
const offlineReady = ref(false)

onMounted(() => {
  window.addEventListener('pwa:update-available', () => { showUpdate.value = true })
  window.addEventListener('pwa:offline-ready', () => { offlineReady.value = true })
})

onUnmounted(() => {
  // cleanup if needed
})

function handleUpdate() {
  applyUpdate()
}

function handleDismiss() {
  showUpdate.value = false
}
</script>

<template>
  <UpdateToast v-if="showUpdate" @reload="handleUpdate" @dismiss="handleDismiss" />
  <!-- existing game UI -->
  <OptimizeDrawer />
</template>
```

`offlineReady` can trigger a one-time toast that says "Ready to play offline" — optional.

4d. `client/services/update.js` — surface connection state (optional companion)

The socket client already auto-reconnects. To give the UI an "offline" indicator, expose the connection state:

```
export const connection = reactive({ state: 'connecting' })
socket.on('connect',   () => { connection.state = 'online' })
socket.on('disconnect', () => { connection.state = 'offline' })
socket.io.on('reconnect_attempt', () => { connection.state = 'reconnecting' })
```

Badge renders as a 8 px dot in the corner: green/yellow/red. Nothing renders in offline mode — the user simply can't start new games, but existing UI elements stay rendered because they're cache-served.

5. Nginx Change — `apps/server.conf`

Add cache headers so the browser doesn't hammer the origin for stale assets:

```
location = /sw.js {
  add_header Cache-Control "no-cache";
}

location ~ /(sw-[^/]+\.js|workbox-[^/]+\.js)$ {
  add_header Cache-Control "no-cache";
}

location = /manifest.webmanifest {
  add_header Cache-Control "no-cache";
}

location ~ /(icons|prism)/ {
  add_header Cache-Control "public, max-age=604800, immutable";
}

# Hashed bundle assets — long-lived
location ~ /assets/.+\.(?:[0-9a-f]{8,}) {
  add_header Cache-Control "public, max-age=31536000, immutable";
}

# Everything else — short default
location / {
  add_header Cache-Control "no-cache";
}
```

Keep the existing location `/ws/` and `/socket.io/` blocks — unchanged.

6. Capacitor Native Build — no regression

- `npm run build:client` emits `dist/client/` and works in Android. The SW file ships inside the APK but **is not registered** (`file://` origin) — expected.
- Workbox will gracefully no-op in `file://` mode.
- The Capacitor WebView is treated as a secure context and can register a SW if needed later.
- Add `"vite-plugin-pwa"` to `capacitor.ignore` in `package.json` if Capacitor's sync tries to pick it up.

7. Acceptance Criteria

- `npm run build:client` emits `dist/client/sw.js` and `dist/client/manifest.webmanifest`.
- Lighthouse PWA audit: installable, offline-capable.
- Chrome DevTools → Application → Service Workers: SW activated, precache lists `index.html`, `app.js`, `styles.css`, `icons.js`, `pwa.js`, `/prism/*`.
- Kill the server → refresh → app loads from cache, UI shows offline state.
- Hit `/api/sessions/*` → network error, not cached.
- Deploy a new build → refresh → "Update" toast appears → tap Reload → new version loads.
- Socket.io connection unaffected throughout.
- Fireboard WS unaffected.
- Lighthouse Accessibility / Best Practices: no regressions.

8. Rollback

- Remove `VitePWA(...)` from `vite.config.js`.
- Ship a kill-switch `sw.js` that calls `self.registration.unregister()` and `self.caches.keys().forall(k=>caches.delete(k))`.
- Revert `client/index.html` meta tags.
- Revert `client/app.js` registration.

9. Follow-Ups

- Capacitor Android wrapper — separate spec.
- Background sync for `/api/sessions/*` mutations — separate spec.
- Push notifications — separate spec.
- Precache game-logic AI for offline play — separate spec.

10. Implementation Order

1. Install `vite-plugin-pwa`.
2. Add `VitePWA(...)` to `vite.config.js`.
3. Add PWA meta tags to `client/index.html`.
4. Add `client/sw.js`.
5. Add `client/app.js` registration.
6. Add `client/components/UpdateToast.vue`.
7. Add `client/services/update.js`.
8. Add `client/components/OptimizeDrawer.vue`.
9. Update `client/App.vue`.
10. Update `apps/server.conf`.
11. Build + deploy.
12. Verify acceptance criteria.

11. Notes

- Workbox's `generateSW` mode is sufficient; `injectManifest` is only needed for custom SW logic.
- Precache manifest is content-hashed; safe to `immutable` on `/assets/*`.
- Socket.io HTTP long-poll fallback must be `NetworkOnly` — caching it breaks the protocol.
- Prism assets are versioned by path (`/prism/1/...`); bump `prism-v1` → `prism-v2` when the sprite sheet changes.
- Tailwind CDN is temporary; a future build-time Tailwind removes this dependency entirely.

Spec 003 — Progressive Web App with Full Offline Assets

Status: Draft — awaiting review **Authored:** 2026-04-16 **Touches:** `client/sw.js`, `client/index.html`, new service worker + manifest, `server/cli.mjs` and `Caddyfile` — no backend changes. Backend changes are limited to serving the Android wrapper

1. Problem

`sw.js` exists but `client/index.html` is a standard SPA: every visit re-fetches `index.html`, the Tailwind CDN (`cdn.tailwindcss.com`), the Google Fonts CSS, and the Phosphor icons. The app loses its value when offline. Goal: on first load, everything that ships with the app is cached and runs offline.

2. Proposed Change

Ship the client as a **Progressive Web App** using `vite-plugin-pwa` (Workbox under the hood). After the first load:

- `index.html`, `manifest.js`, `app.js`, `icons.js`, `pwa.js`, `styles.css`, and `/prism/*` assets are precached and served from cache-first.
- `sessions.js` traffic to `/api/sessions/*` — must not be cached, ever.
- `/ws/pty` and `/ws/sessions/*` are WebSocket — unaffected by the service worker.
- Third-party scripts (`jsq`, Firebase) run fetch-first or bypass the service worker.
- Fresh client visits **update prompt** appears when a newer build is deployed; user taps to reload into the new version.
- **Web app manifest** with `start_url: "/"`, `display: "standalone"`, icons for Android/iOS and splash screens.
- **Offline indicator** — a small UI hint when `navigator.onLine` is false.

2a. Scope — what ships

- Vite dev dependency: `vite-plugin-pwa` (peer: `vite ^5`).
- Socket protocol, game logic, AI scoring — unchanged. No server-side gameplay changes.
- Capacitor Android wrapper — out of scope for this spec. Will consume `dist/client/` later; the PWA must work inside a Capacitor WebView as a baseline test.
- Push notifications — out of scope. Separate spec.
- Background sync — out of scope. Queue-and-retry for `/api/sessions/*` is a follow-up.

2b. Scope — what stays the same

- Single-process, game-logic AI, scoring — unchanged.
- Caddy reverse proxy, TLS, `/ws/*` WebSocket upgrade — unchanged. Caddy must additionally serve `/sw.js` and `/manifest.webmanifest` with `Cache-Control: no-cache` so updates propagate.
- Authentication — none. Deployment remains behind Caddy basic auth.
- Fireboard integration — unchanged. Its WS traffic bypasses the service worker entirely.

3. Build-Time Changes

3a. Dependency

```
npm i -D vite-plugin-pwa
```

`vite-plugin-pwa` ships with Workbox, no additional runtime bundle other than the service worker itself (~8 KB) and serves `vite ^5` without conflict.

3b. `vite.config.js`

```
import { defineConfig } from 'vite'
import vue from '@vitejs/plugin-vue'
import { VitePWA } from 'vite-plugin-pwa'
import { cloneDb } from 'vite-plugin-pwa'

export default defineConfig({
  plugins: [vue(),
    VitePWA({
      mode: 'production',
      strategies: 'generateSW',
      registerType: 'prompt',
      injectRegister: false,
      manifest: false,
      workbox:{
        globPatterns:['**/*.{js,css,html,ico,svg,woff2}'],
        maximumFileSizeToCacheInBytes: 5242880,
        navigateFallback: null,
        cleanupOutdatedCaches: true,
        clientsClaim: false,
        skipWaiting: false,
        runtimeCaching:[
          // Socket.io HTTP long-poll fallback
          {urlPattern:/^\/socket\.io\/.*$/,handler:'NetworkOnly',method:'GET'},
          {urlPattern:/^\/api\/sessions\/.*$/,handler:'NetworkOnly',method:'GET'},
          {urlPattern:/^\/api\/.*$/,handler:'NetworkOnly',method:'GET'},
          {urlPattern:/^\/prism\/.*$/,handler:'CacheFirst',options:{cacheName:'prism-v1',expiration:{maxEntries:30,maxAgeSeconds:604800}}},
          {urlPattern:/^https://cdn\.tailwindcss\.com/.*$/,handler:'StaleWhileRevalidate',options:{cacheName:'tailwind-cdn',expiration:{maxEntries:5}}},
          {urlPattern:/^https://fonts\.(googleapis|gstatic)\.com/.*$/,handler:'StaleWhileRevalidate',options:{cacheName:'google-fonts',expiration:{maxEntries:20,maxAgeSeconds:31536000}}},
          {urlPattern:/^https://unpkg\.com/phosphor-icons/.*$/,handler:'StaleWhileRevalidate',options:{cacheName:'phosphor-icons',expiration:{maxEntries:5,maxAgeSeconds:604800}}},
          {urlPattern:/^https?://(?:[0-9]{1,3}\.){3}[0-9]{1,3}(?:[:/].*)?$/,handler:'NetworkOnly'},
          {urlPattern:/^https?:\/\/[^/]+\/(?:socket\.io|api\/sessions|api|prism)(?:[/?].*)?$/,handler:'NetworkOnly'},
          {urlPattern:/^https?:\/\/[^/]+\/(?:[/?].*)?$/,handler:'NetworkFirst',options:{cacheName:'app-shell',networkTimeout:3,expiration:{maxEntries:60,maxAgeSeconds:86400}}},
          {urlPattern:/^https?:\/\/[^/]+\/(?:[/?].*)?$/,handler:'NetworkOnly'},
        ],
        cleanup:[{name:'prism-v0'},{name:'tailwind-cdn'},{name:'google-fonts'},{name:'phosphor-icons'},{name:'app-shell'}],
        manifestTransforms:[(m,b)=>{
          const d=cloneDb(b);
          for(e of d){
            if(/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i.test(e.name)){e.revision=null;continue}
            if(!e.revision&&/^\/(?:[^/]+\.(?:[cm]?[jt]s|[jt]sx?|[a-z0-9]+\.[a-z0-9]+)(?:[?#].*)?)?$/i.test(e.name)){e.revision=null}
          }
          return{manifest:m,warnings:[]}
        }],
        globIgnores:[],
        dontCacheBustURLsMatching:/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i,
        modifyURLPrefix:{'':'/'},
        navigationPreload:false,
        sourcemap:false,
        maximumFileSizeToCacheInBytes:5242880,
        mode:'production',
        swDest:'dist/client/sw.js',
        globDirectory:'dist/client',
        rollupFormat:'iife',
        minify:true,
        inlineWorkerRuntime:true,
        inlineSocketIoPolling:true,
        cleanupOutdatedCaches:true,
        clientsClaim:false,
        skipWaiting:false,
        navigationPreload:false,
        navigationFallback:null,
        globPatterns:['**/*.{js,css,html,ico,svg,woff2}'],
        globIgnores:[],
        dontCacheBustURLsMatching:/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i,
        manifestTransforms:[(m,b)=>{...}],
        modifyURLPrefix:{'':'/'},
        maximumFileSizeToCacheInBytes:5242880,
        cleanup:[...],
        runtimeCaching:[...],
        navigationFallback:null,
        navigationFallbackDenylist:[...],
        navigationFallbackAllowlist:[...],
        sourceMap:false,
        dev:{...},
        base:'/',
        scope:'/',
        strategies:'generateSW',
        registerType:'prompt',
        injectRegister:false,
        manifest:false,
        inlineManifestLinkIntoHtml:false,
        injectManifest:false,
        injectManifestPlugin:false,
        devOptions:{...},
        buildOptions:{...},
        pluginVersion:'0.x.x',
        vitePluginVersion:'0.x.x',
        viteVersion:'5.x.x',
        workboxVersion:'7.x.x',
        generatedAt:'2026-04-16T...',
        ...
      },
      devOptions:{
        enabled: false, // do not run SW in dev — HMR and SW conflict
        type: 'classic',
      },
      includeAssets:[],
      filename:'sw.js',
      outDir:'dist/client',
      srcDir:'.',
      publicDir:'public',
      base:'/',
      scope:'/',
      type:'classic',
      swDest:'dist/client/sw.js',
      globDirectory:'dist/client',
      rollupFormat:'iife',
      minify:true,
      inlineWorkerRuntime:true,
      inlineSocketIoPolling:true,
      cleanupOutdatedCaches:true,
      clientsClaim:false,
      skipWaiting:false,
      navigationPreload:false,
      navigationFallback:null,
      globPatterns:['**/*.{js,css,html,ico,svg,woff2}'],
      globIgnores:[],
      dontCacheBustURLsMatching:/^\/(?:prism\/.*|icons\/.*|[^/]+\.(?:ico|svg|woff2?|ttf|otf))$/i,
      manifestTransforms:[(m,b)=>{...}],
      modifyURLPrefix:{'':'/'},
      maximumFileSizeToCacheInBytes:5242880,
      cleanup:[...],
      runtimeCaching:[...],
      navigationFallback:null,
      navigationFallbackDenylist:[...],
      navigationFallbackAllowlist:[...],
      sourceMap:false,
      dev:{...},
      base:'/',
      scope:'/',
      strategies:'generateSW',
      registerType:'prompt',
      injectRegister:false,
      manifest:false,
      inlineManifestLinkIntoHtml:false,
      injectManifest:false,
      injectManifestPlugin:false,
      devOptions:{...},
      buildOptions:{...},
      pluginVersion:'0.x.x',
      vitePluginVersion:'0.x.x',
      viteVersion:'5.x.x',
      workboxVersion:'7.x.x',
      generatedAt:'2026-04-16T...',
      ...
    })
  ],
  server:{
    port:5173,
    strictPort:false,
    host:true,
    proxy:{
      '/socket.io':{target:'http://localhost:3000',ws:true,changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/api':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/ws':{target:'http://localhost:3000',ws:true,changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/prism':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/icons':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/manifest.webmanifest':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/sw.js':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js':{target:'http://localhost:3000',changeOrigin:false,secure:false,rewrite:null,configure:null,bypass:null,logLevel:'info'},
      '/workbox-*.js.map':{target:'http://localhost:3000',changeOrigin:false,secure
```